

IMF2 Locking: $\ln(1+\alpha z)$ Axis Comparison

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Dataset: bao_hht_2bin_z0p7_longbest • IMF: 2 (IMF2) • Method(s): CEEMDAN • pads: 0.12 • ngrid: 384 • seeds: 3,7,11

Executive summary

Paired comparison between $\alpha=0.5$ (FR-style $\ln(1+\frac{1}{2}z)$) and $\alpha=1.0$ (standard $\ln(1+z)$) shows $\alpha=0.5$ wins on $|r|$ in 0.0% of runs and achieves lower permutation p in 0.0% of runs. Mean $|r|$: $\alpha=0.5=0.8791$, $\alpha=1.0=0.9094$. Median p : $\alpha=0.5=0.004975$, $\alpha=1.0=0.004975$.

Why α matters

The transform $\ln(1+\alpha z)$ re-parameterizes the redshift axis used to build a uniform analysis grid. $\alpha=0.5$ compresses high- z relative to $\alpha=1.0$, which can stabilize instantaneous-frequency estimates in CEEMDAN/EMD decomposition, potentially improving phase-locking detection between DH_over_rd and DM_over_rd (IMF2).

Relation to PLamb FR/Chirp expectations

PLamb's FR conjecture suggests an effective scaling closer to $\ln(1+\frac{1}{2}z)$ may better linearize the latent 'chirp' in residuals, if the acceleration signature accumulates with look-back in a sub-linear fashion. If $\alpha=0.5$ produces systematically larger $|r|$ and smaller p (and especially lower BH q -values), it supports the view that the IMF2 structure is more coherent under FR-style scaling.

By-method snapshot

method	$\alpha=0.5$ wins ($ r $)	$\alpha=0.5$ wins (p)
ceemdan	0.0%	0.0%

Caveats

- Small-sample (7 z -nodes) and finite-sample CEEMDAN variance means wins should replicate across seeds, pads, and grids.
- Surrogate p -values reflect phase-randomized null on DH; alternative nulls (block-shuffle, amplitude-adjusted surrogates) may be explored.
- Interpret α -wins in concert with BH-adjusted q -values and stability across hyperparameters.

Repro notes

Inputs: locking_alpha0p5.csv, locking_alpha1p0.csv

Paired CSV: locking_alpha_compare.csv

Generated by hht_alpha_compare_report.py